Performance Evaluation of FMCW Radar with CFAR Detection under Low-SNR and Clutter Conditions

1. Introduction

Frequency Modulated Continuous Wave (FMCW) radar is widely used in automotive radar, surveillance systems, drone detection, and defense applications. This project focuses on the simulation and performance analysis of an advanced FMCW radar system using signal processing techniques in Python.

The objective of the project is to design and simulate a complete radar signal chain including signal generation, target modeling, range detection, velocity estimation, multi-target detection, noise suppression, and tracking.

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2. Objectives

The main objectives of this project are:

- To simulate FMCW chirp signal generation
- To model single and multiple moving targets
- To implement Range FFT for distance estimation
- To implement Doppler FFT for velocity estimation
- To generate Range-Doppler maps
- To apply CFAR (Constant False Alarm Rate) detection
- To evaluate detection performance using ROC curves
- To implement Kalman filtering for target tracking
- To analyze Radar Cross Section (RCS) impact

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3. FMCW Radar Principle

An FMCW radar transmits a chirp signal whose frequency increases linearly with time. When the transmitted signal reflects from a target, it returns with a time delay. The frequency difference (beat frequency) between transmitted and received signals is proportional to the target's range.

Range Equation:

$$R = \frac{c \cdot f_b}{2S}$$

Where:

- c= Speed of light
- f_b= Beat frequency
- S= Chirp slope

Velocity is determined using Doppler frequency shift extracted through 2D FFT processing.

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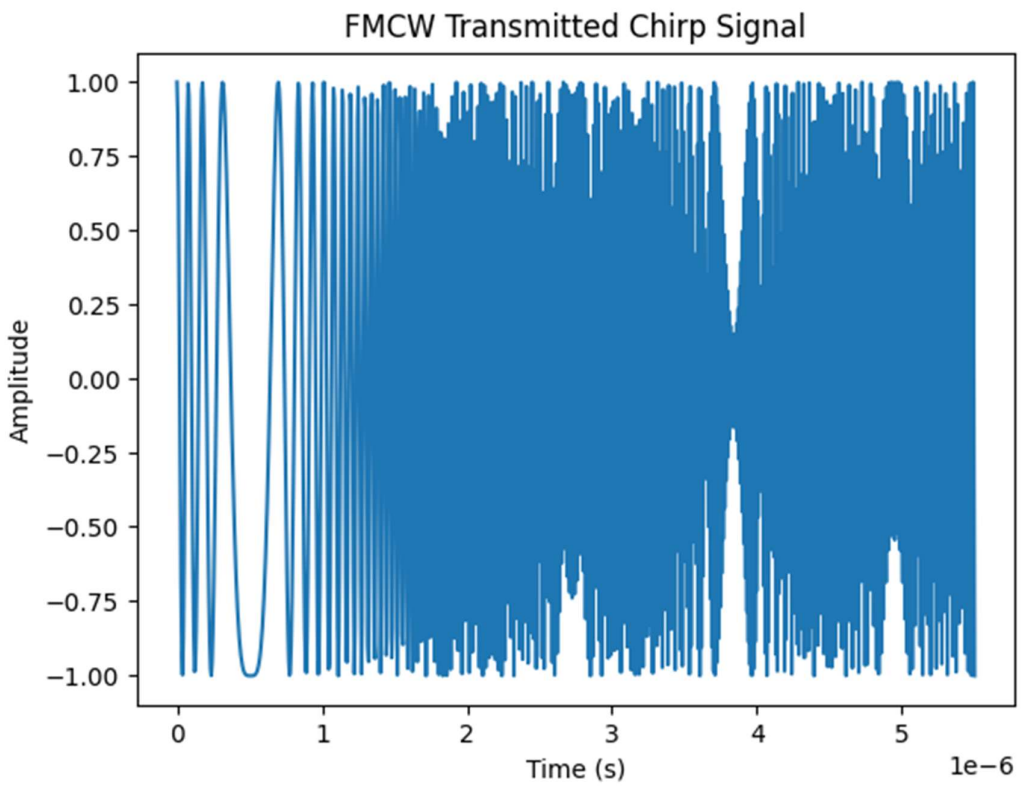
4. System Design and Implementation

4.1 Chirp Signal Generation

A linear frequency modulated signal was generated using defined parameters such as:

- Carrier Frequency
- Bandwidth
- Chirp Time
- Sampling Frequency

Transmitted Chirp Signal Plot:

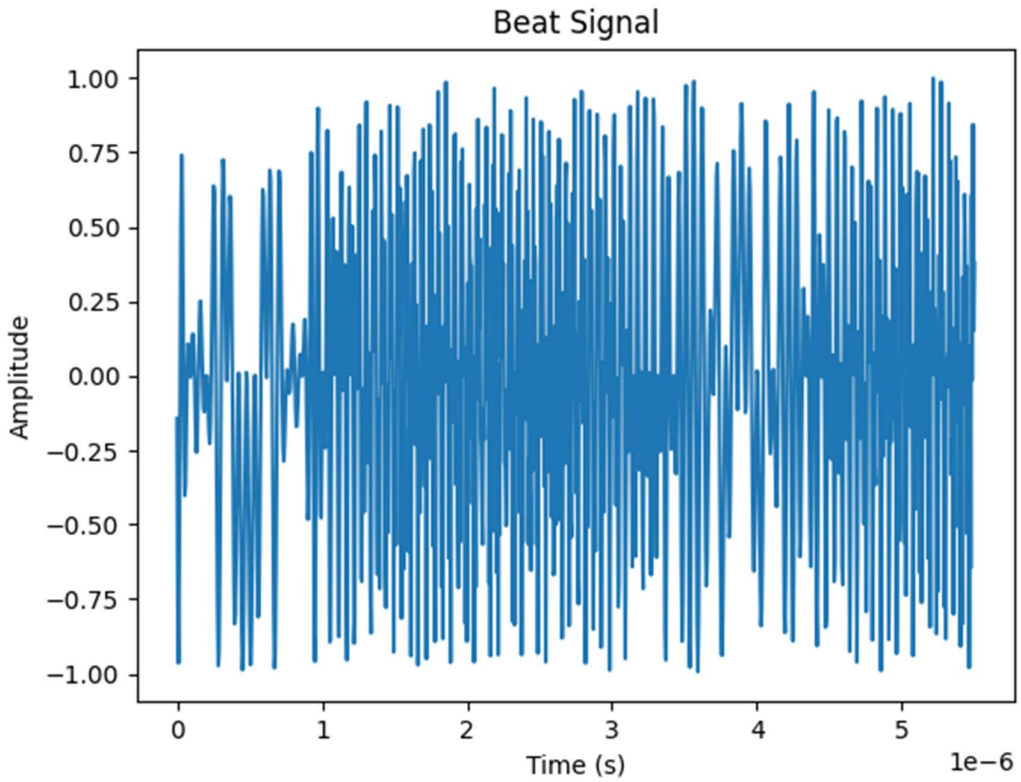


4.2 Target Modeling

Single and multiple targets were simulated with:

- Different ranges
- Different velocities
- Additive White Gaussian Noise (AWGN)

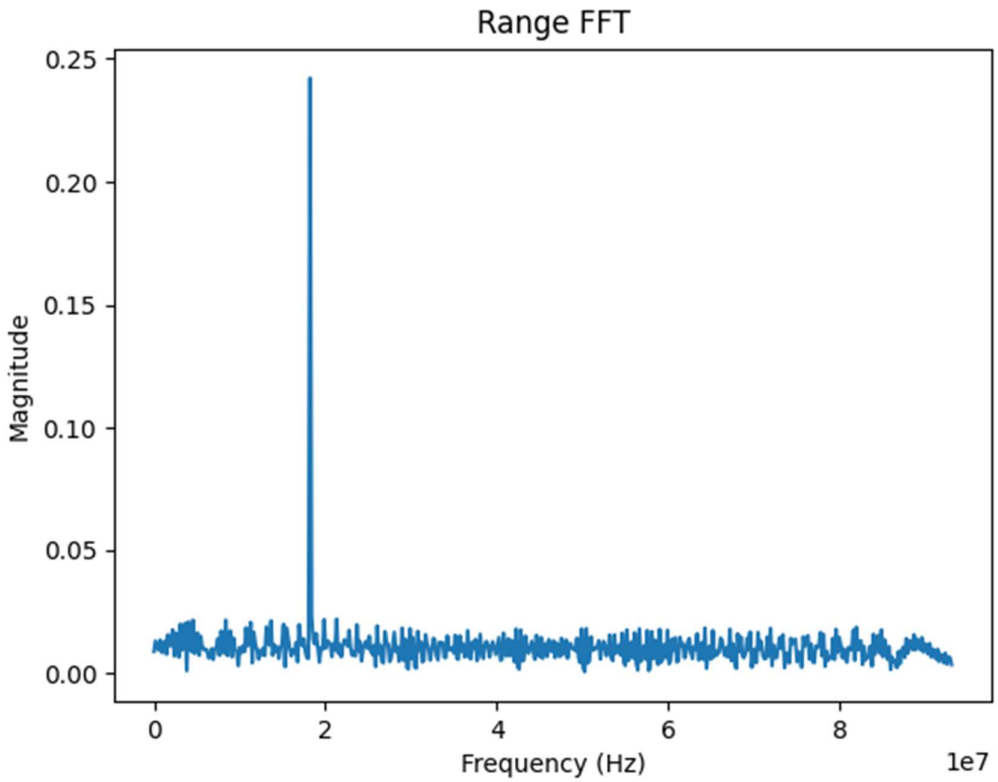
Time Delay Simulation image:



4.3 Range Estimation – FFT

Fast Fourier Transform (FFT) was applied to extract beat frequency components corresponding to target distance.

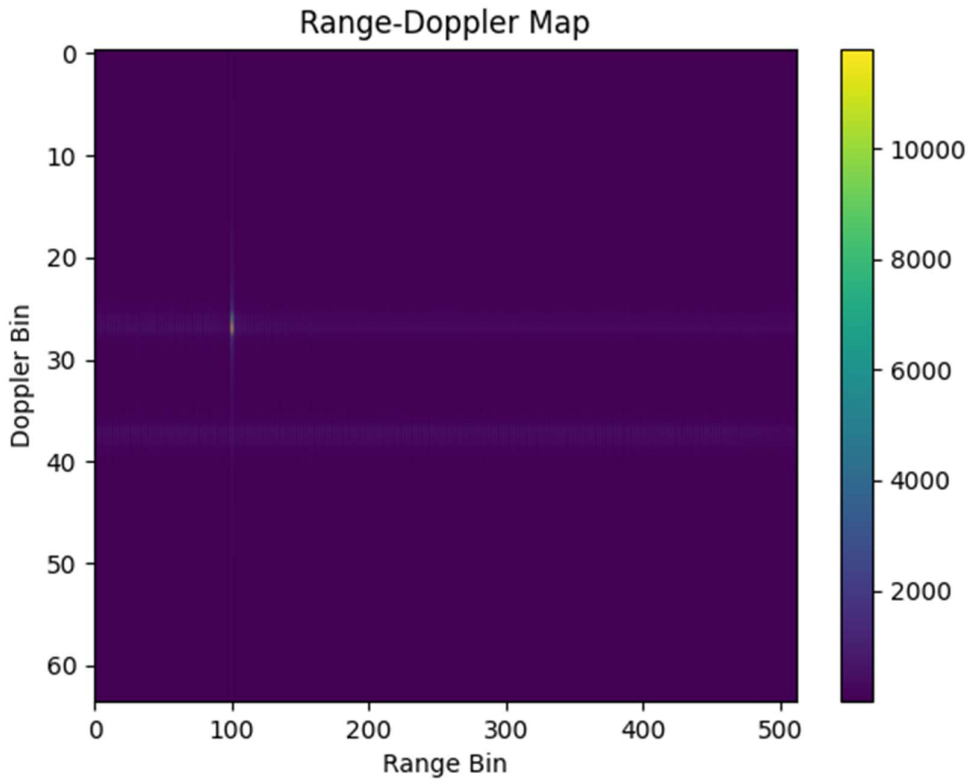
Range FFT Plot:



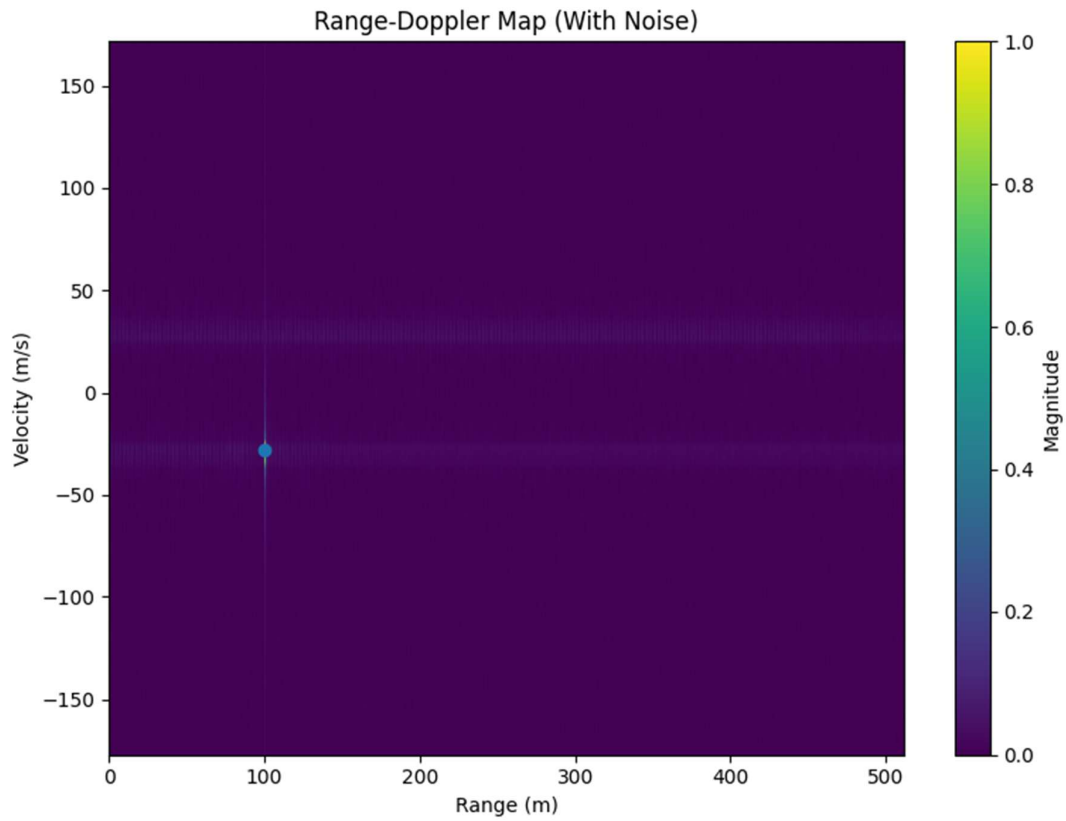
4.4 Range-Doppler Map

A 2D FFT was implemented to extract both range and velocity information, producing a Range-Doppler Map.

Range-Doppler Map Image:



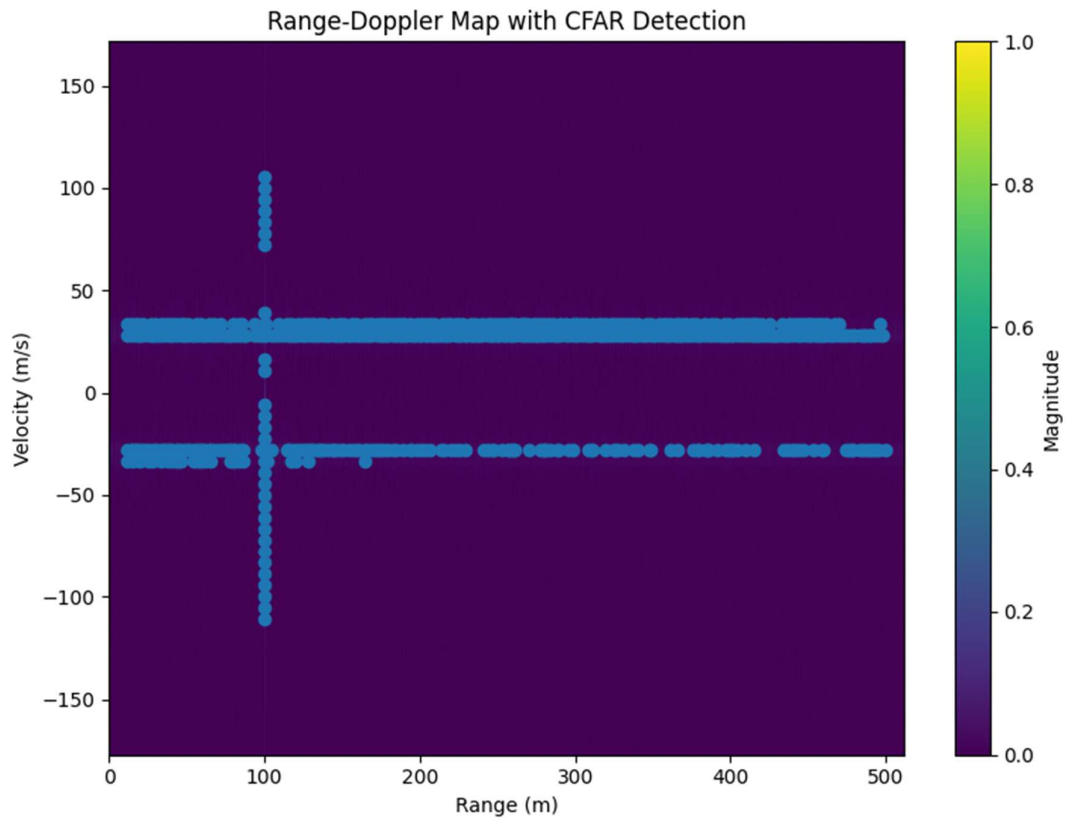
Range-Doppler map (with noise):



4.5 CFAR Detection

To improve detection reliability in noisy environments, a CFAR algorithm was implemented. It adaptively sets threshold levels based on surrounding noise cells to reduce false alarms.

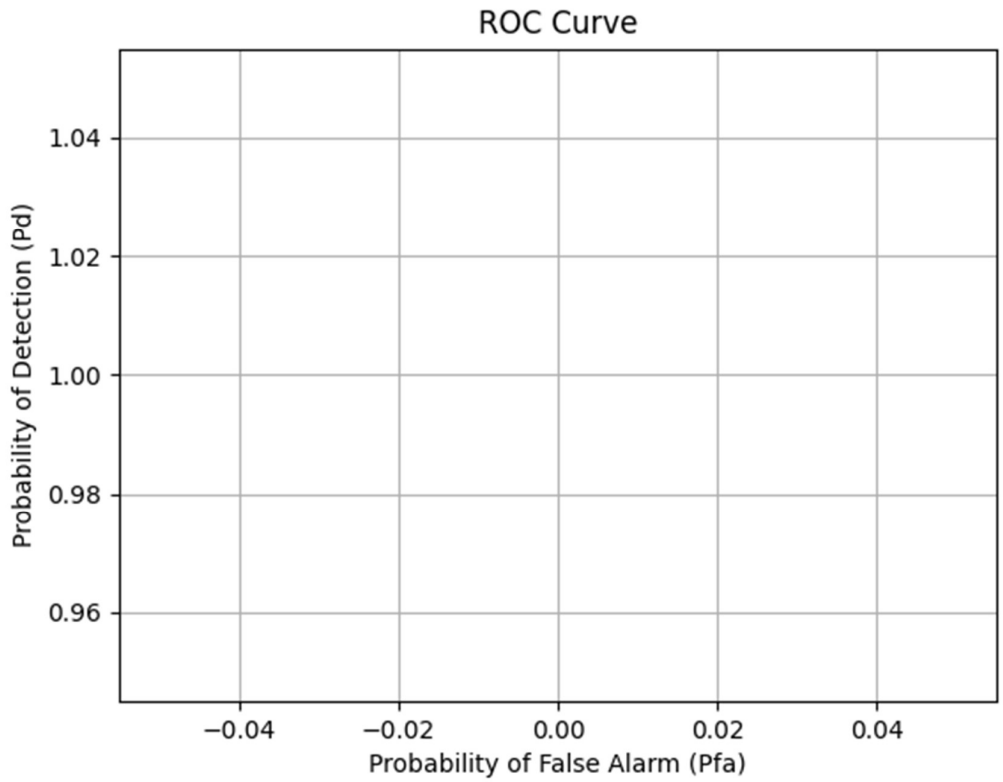
CFAR Output Image:



4.6 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curve was generated to analyze detection probability versus false alarm rate.

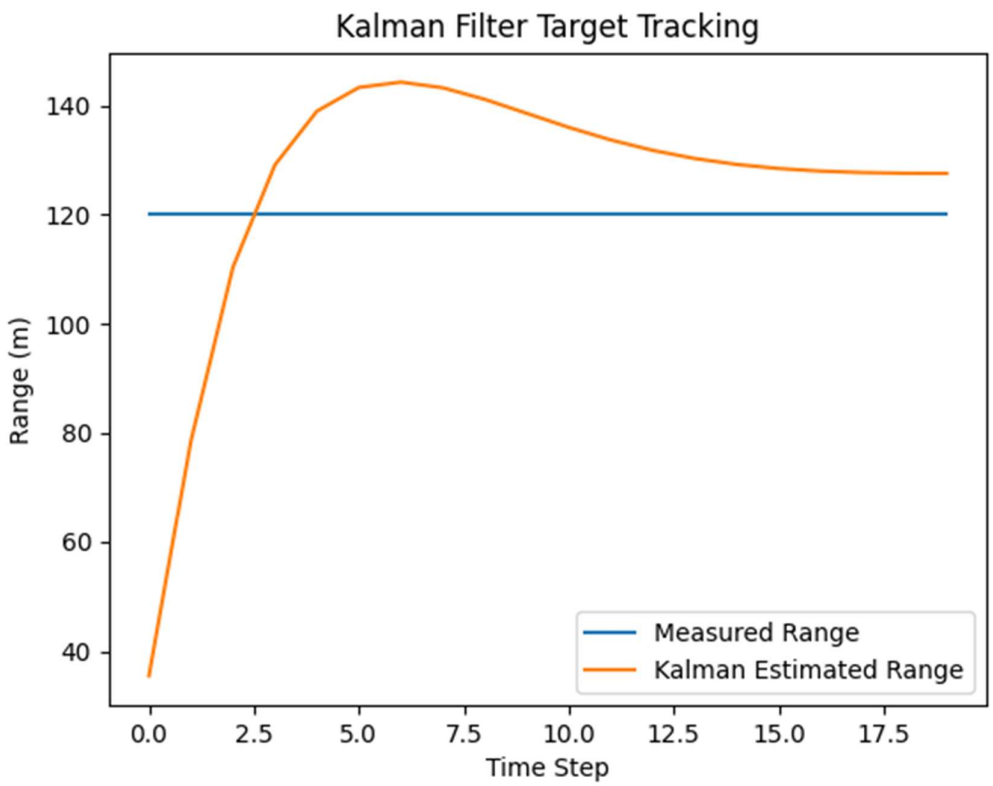
ROC Curve Image:



4.7 Kalman Filter Tracking

A Kalman Filter was implemented for tracking moving targets. It improves estimation accuracy by predicting future states and correcting based on measurements.

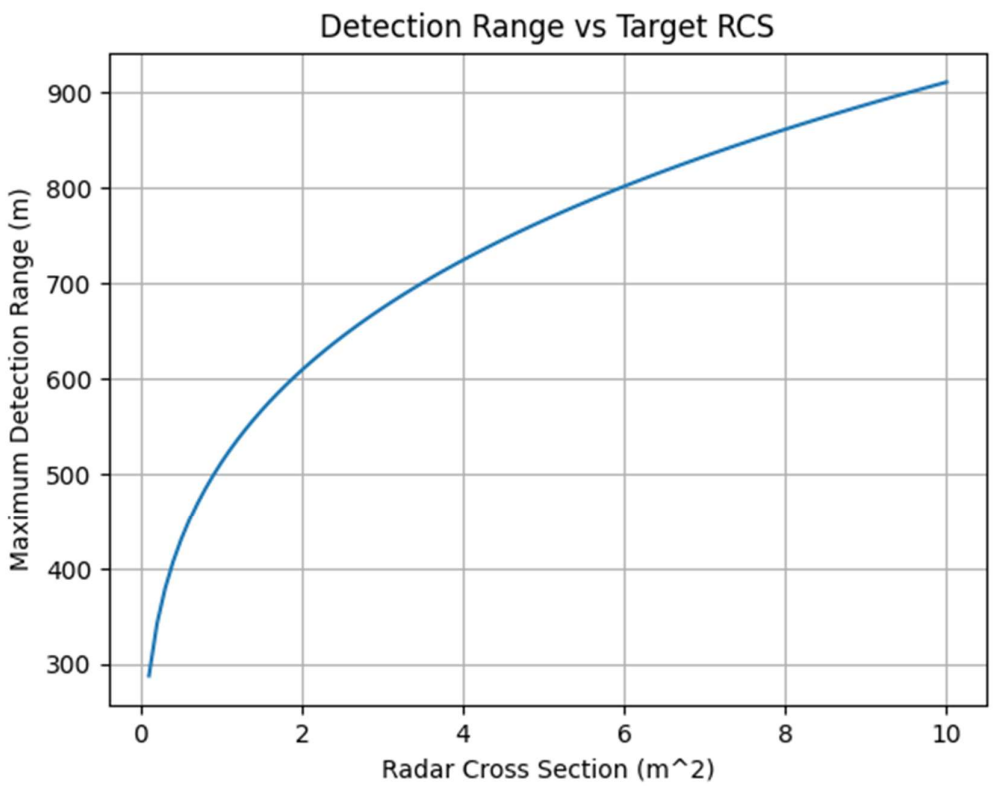
Tracking Output Plot:



4.8 Radar Cross Section (RCS) Analysis

The effect of target reflectivity (RCS) on detection capability was studied to understand how radar performance varies with target size and material properties.

RCS Analysis Plot:



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5. Results and Observations

- Accurate range estimation was achieved using FFT.
- Range-Doppler mapping successfully detected moving targets.
- CFAR significantly reduced false alarms in noisy environments.
- ROC curve demonstrated improved detection probability with optimized thresholding.
- Kalman filter enhanced tracking stability and prediction accuracy.
- Multi-target detection was successfully simulated.

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6. Tools and Technologies Used

- Python
- NumPy
- SciPy

- Matplotlib
- Signal Processing Techniques
- FFT Algorithms
- Statistical Detection Methods

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7. Applications

- Automotive Radar Systems
- Drone Detection
- Military Surveillance
- Air Traffic Monitoring
- Collision Avoidance Systems

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8. Conclusion

This project successfully demonstrates the complete signal processing chain of an FMCW radar system through simulation. Advanced techniques such as CFAR detection, ROC analysis, and Kalman filtering enhance the practical relevance of the system. The implementation reflects a strong understanding of radar signal processing concepts and real-world radar system behavior.

The project can be further extended using real hardware platforms such as SDR or FPGA-based implementations for real-time deployment.